

Comparative Evaluation of Deep Learning Architectures and Training Strategies for Diabetic Retinopathy Classification

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MOTIVATION

Diabetic retinopathy (DR) is a leading cause of preventable blindness. Deep learning on retinal fundus photographs offers scalable automated screening — but most published studies evaluate a single architecture on a single dataset with non-standardized protocols. This fragments the field, makes cross-method comparison unreliable, and hides the real, practical impact of transfer learning versus training from scratch.

CONTRIBUTIONS

- A standardized evaluation framework enabling fair, reproducible comparison across 4 heterogeneous public datasets under identical experimental conditions
- The first systematic cross-dataset replication study of 6 literature methods, exposing performance variability that single-benchmark reporting hides
- A controlled ablation of ImageNet transfer learning vs. training from scratch across 4 architectures — 64 conditions in total
- Evidence-based, actionable guidance: prioritize multi-dataset robustness over single-benchmark peak accuracy when designing clinical screening systems

DATASETS EVALUATED

Dataset	Size	Max Resolution	Country
APTOS 2019	5,590	3216×2136	India
IDRID	516 / 81 / 516	4288×2848	India
DDR	13,673	3216×2136	China
Messidor-2	1,748	2304×1536	France

Chosen to span device type, population, resolution, and class balance — a stress test that single-dataset evaluations cannot expose.

CLASSIFICATION TASKS

- Binary — No DR vs. Any DR (screening referral), and Non-proliferative vs. Proliferative DR (urgent intervention)
- Multiclass — 5 severity grades, No DR through Proliferative, for fine-grained treatment/progression monitoring
- Multiclass is substantially harder: subtle differences between adjacent grades (e.g., Moderate vs. Severe) drive large accuracy drops

DETAILED RESULTS — AVERAGED ACROSS ALL FOUR DATASETS

Experiment 1 — Method Averages (Binary & Multiclass Accuracy)

Method	Avg. Binary Acc.	Binary σ	Avg. Multiclass Acc.
Modified Xception	0.752	0.166	0.582
Hybrid CNN-SVD + ELM	0.759	0.114	0.592
Hybrid MobileNetV2 + SVM	0.856	0.070	0.699
EfficientNet-B5 + CLAHE	0.702	0.060	0.557
VGG16 + novel preprocessing	0.747	0.106	0.572
VGG16	0.775	0.081	0.602
InceptionV3	0.783	0.058	0.616

A low standard deviation alone is not sufficient: EfficientNet-B5+CLAHE is the most "consistent" by σ yet fails clinically (near-zero specificity on IDRiD). MobileNetV2+SVM pairs a high mean with low variance and balanced precision/recall — the combination that matters for deployment.

EVALUATION FRAMEWORK



Preprocessing: resizing/cropping, optic-disc centering, channel normalization, CLAHE & gamma correction, artifact masking. Augmentation: flips, rotation, translation/scale, brightness/contrast jitter, constrained elastic transforms. Metrics: Accuracy, Precision, Recall, F1, Specificity, AUROC, Quadratic Weighted Kappa.

EXP. 1 — LITERATURE REPLICATION

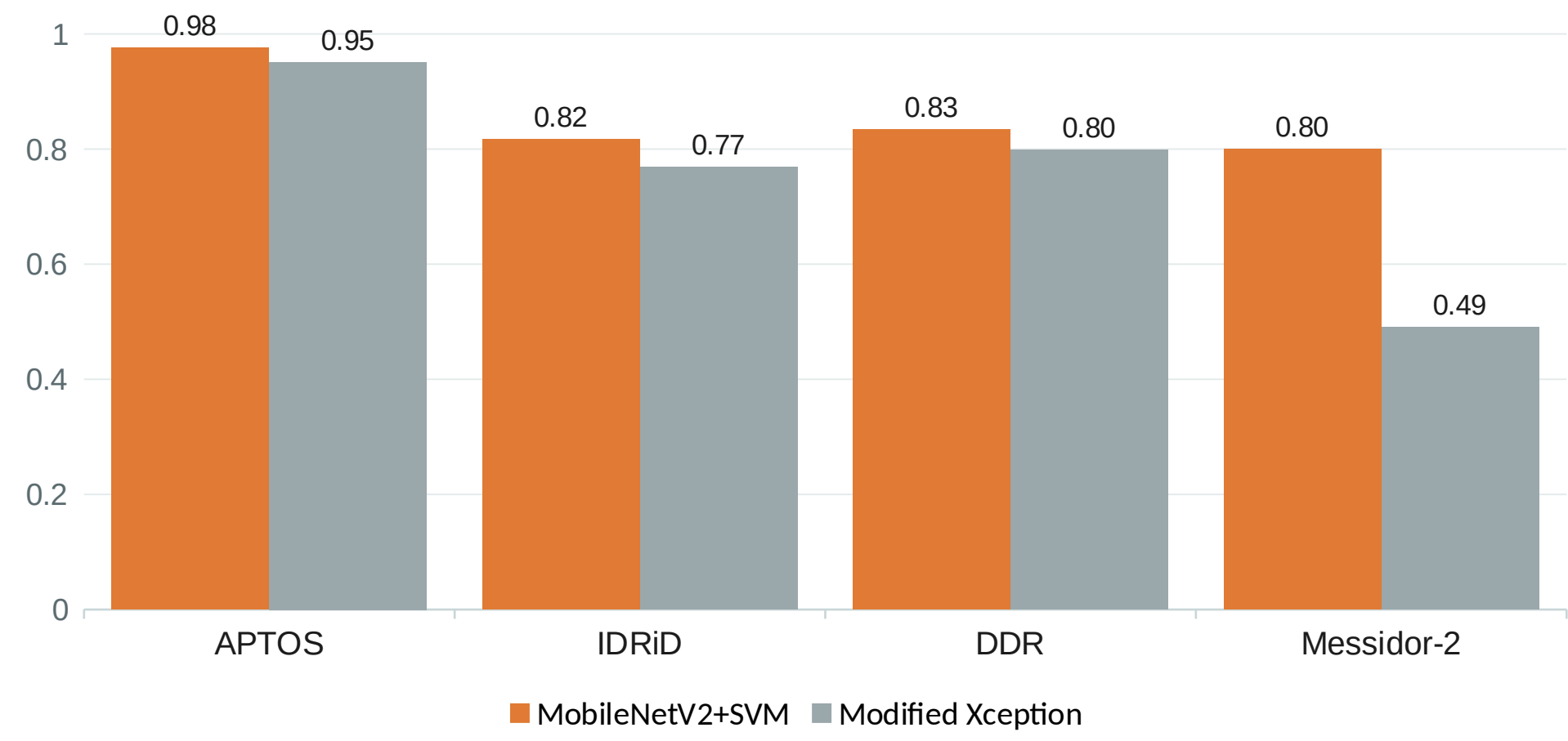
Six representative methods, spanning every major architectural paradigm in the DR literature, retrained under identical splits on all four datasets:

- Modified Xception — deep layer aggregation + Grad-CAM localization
- Hybrid CNN-SVD + ELM — SVD feature reduction, fast convergence
- Hybrid MobileNetV2 + SVM — lightweight CNN features, SVM classifier
- EfficientNet-B5 + CLAHE — compound scaling, contrast enhancement
- VGG16 + novel color-based preprocessing layer
- VGG16 & InceptionV3 — standard transfer-learning baselines

EXP. 2 — TRANSFER LEARNING ABLATION

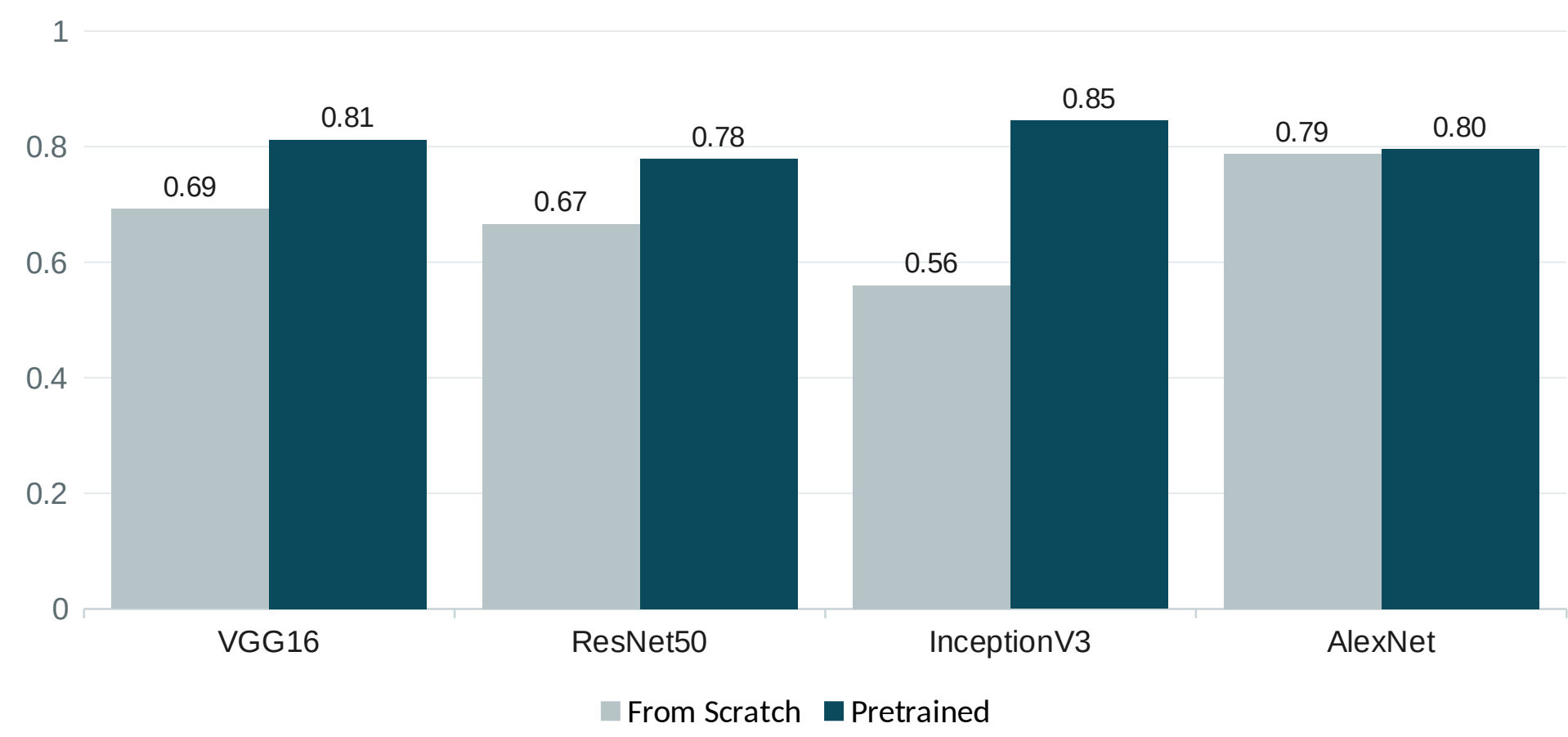
- 4 architectures compared: VGG16, ResNet50, InceptionV3, AlexNet
- From-scratch initialization vs. ImageNet-pretrained initialization
- Identical 70/15/15 splits (seed 42), 224×224 input, Adam optimizer, cross-entropy loss, batch size 32, early stopping (patience 20)
- 64 total conditions: 4 architectures × 4 datasets × 2 tasks × 2 training strategies

CROSS-DATASET CONSISTENCY



Hybrid MobileNetV2+SVM sustains ≥ 0.78 binary accuracy on every dataset (cross-dataset $\sigma \approx 0.07$). Modified Xception swings from 0.951 (APTOS) to 0.491 (Messidor-2) — architectural sophistication does not guarantee robustness.

TRANSFER LEARNING IMPACT



Average binary accuracy, from-scratch vs. ImageNet-pretrained, across all 4 datasets. Pretraining wins in every one of 32 binary configurations tested (avg. +18.7%); deeper InceptionV3 gains most (+46 pts on APTOS), simpler VGG16 least (+1 pt).

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multiclass conditions where from-scratch beat pretrained

20–30%

cross-dataset accuracy swing — a clinical deployment risk

8.4×

fewer parameters: MobileNetV2 vs. ResNet50

KEY FINDINGS

- Sophistication $\neq$ robustness: Xception and EfficientNet-B5+CLAHE degrade sharply on IDRiD/DDR despite strong single-dataset scores
- ImageNet transfer learning is essential, not optional, for limited medical-imaging data — use it regardless of architecture
- Hybrid deep-feature + classical-classifier models (MobileNetV2+SVM) generalize best, especially for multiclass severity grading

CONCLUSION & FUTURE WORK

- Single-dataset benchmark accuracy (e.g. 0.95 on APTOS) overstates real-world utility — local validation before clinical deployment is required
- Future work: broader architecture/dataset coverage, cross-device domain adaptation, explainability & fairness audits, edge-hardware latency optimization

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